

January 26, 2026, second midterm, Computer & Network Security

SURNAME: _____ NAME: _____ MATRICOLA: _____

Q1: Multiple answer questions (comments can be added on the right if/when necessary)

In the Boneh–Franklin Identity-Based Encryption scheme, the PKG:	• Certifies Identities • Certifies Public Keys associated to the Identities • Can encrypt any message exchanged by any pair of users • Can decrypt any message exchanged by any pair of users
In a “vanilla” ElGamal encryption scheme defined over a cyclic group whose elements are represented using 4096 bits, what is the size of the ciphertext (all included) when encrypting a message of 1800 bits?	• 1800 bits • 2048 bits (the smaller power of 2 superior to the msg size) • 3600 bits • 4096 bits • 8192 bits
$P, Q \rightarrow$ points of additive EC group (order p). Let $\text{COM}(m, k) = [m]P + [k]Q$ be a Pedersen commitment for message m using randomness k . Which relation between points P and Q , once known, breaks the binding property?	• Find a such that $P = [a]Q$ • Find a such that $Q = [a]P$ • Find a such that $P = Q^a \text{ mod } p$ • Find a such that $Q = P^a \text{ mod } p$
Consider $\mathbb{Z}_{23}^*$ and a point a of the group. Which of the following statements are correct? (can be more than one).	• $a=5$ is a quadratic residue mod 23 • $a=5$ is NOT a quadratic residue mod 23 • a^{22} is equal to 1 if and only if a is a quadratic residue mod 23 • a^{22} is equal to 1 if and only if a is NOT a QR mod 23 • if a^{11} is equal to 1, then a is a quadratic residue mod 23 • if a^{11} is equal to 1, then a is a NOT a QR mod 23
In the Joux 3-way Diffie-Hellman construction based on a pairing $e(G, G) \rightarrow G_t$ – being P_x the public coefficient exchanged by party x , and S the resulting shared secret:	• P_x and S are both points of group G • P_x and S are both points of group G_t • P_x is a point of group G, while S is a point of group G_t • P_x is a point of group G_t, while S is a point of group G

Q2. Consider the Elliptic curve $y^2 = x^3 + 2x + 3$ defined over the modular integer field $\mathbb{Z}_7$.

A. find all the points $EC(\mathbb{Z}_7)$

(2,1) (2,6) (3,1) (3,6) (6,0) O

$$\begin{aligned}
 P &= (x_1, y_1) \\
 Q &= (x_2, y_2) \\
 R &= P + Q = (x_3, y_3) \\
 x_3 &= \lambda^2 - x_1 - x_2 \\
 y_3 &= \lambda(x_1 - x_3) - y_1 \\
 \lambda &= \begin{cases} \frac{y_2 - y_1}{x_2 - x_1} & P \neq Q \\ \frac{3x_1^2 + a}{2y_1} & P = Q \end{cases}
 \end{aligned}$$

B. Is there a point in the group (other than the identity) which is **equal to its inverse**? If yes, determine it.

(6,0)

C. Compute $(2,1) + (2,1)$ detailing the specific step-by-step computations [possibly useful formulas are here on the right]

(3,6)

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Q3. Assume arithmetic modulus 7. A Linear secret sharing scheme involving 3 parties is described by the following access control matrix:

A:	1	2	3
B:	0	5	0
C:	0	0	4

What is the secret S, assuming that the following shares are revealed:

A $\rightarrow$ 4
B $\rightarrow$ 3
C $\rightarrow$ 5

Careful with modular inverses! $S = A - B \times 2 \times 5^{-1} - C \times 3 \times 4^{-1} = A - 6B - 6C = 5$

Q4 Describe the Boneh-Franklin Identity Based Encryption scheme

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Q5 - A (2,5) Shamir Secret Sharing scheme uses a non-prime modulus $p=55$ (if you need modular inverses see table on the right). Of the 5 participating parties $P_1, \dots, P_5$, with respective x coordinates $x_i = \{1, 2, 3, 4, 5\}$, parties P_3 , and P_5 aim at reconstructing the secret.

a) Compute the Lagrange Interpolation coefficients for parties 3 and 5.

b) Reconstruct the secret, assuming that the shares are:

$$P_3 \rightarrow 35$$

$$P_5 \rightarrow 51$$

c) Does the knowledge of share P_3 leak information about the secret?

d) Does the knowledge of share P_5 leak information about the secret?

,Lagrange coeff: 30, 26.

Secret: $30 \times 35 + 26 \times 51 \bmod 55 = 11$

P_3 does not leak, while P_5 does

(why? Because share = secret + random $x_i \bmod 55$

$\rightarrow$ with $x_i=5$, share not anymore random ($5 \times \text{random} \bmod 55$ is not random: only 11 values)

x	1/x mod 55
1	1
2	28
3	37
4	14
6	46
7	8
8	7
9	49
12	23
13	17
14	4
16	31
17	13
18	52
19	29
21	21
23	12
24	39
26	36
27	53
28	2
29	19
31	16
32	43
34	34
36	26
37	3
38	42
39	24
41	51
42	38
43	32
46	6
47	48
48	47
49	9
51	41
52	18
53	27
54	54

Q6. In Threshold RSA, how could Victor Shoup's technique extract the digital signature $H(m)^d$ from the previously computed term $H(m)^{L!d}$?

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Q7 A message M , which you know is either M_1 or M_2 , is encrypted using El Gamal over an Elliptic Curve group of prime order p , where g is the generator and $h=g^s$ is the public key of the recipient – with multiplicative notation:

$$\text{ENC}(M) = (R, c) = (g^r, M h^r)$$

Assume now that the curve admits bilinear pairing, and **show how the existence of the pairing allows one to break the semantic security of El Gamal and determine whether the encrypted message is M_1 or M_2 , without knowledge of the secret key**

Pick one among the msgs, e.g. M_1 , construct $X=c M_1^{-1} = M h^r/M_1$: if $M=M_1$ this becomes $h^r = g^{rs}$
Then break DDH using pairing, i.e. check if $e(g, X) =? e(g^r, h)$